NeuralPlane — Structured 3D Reconstruction in Planar Primitives

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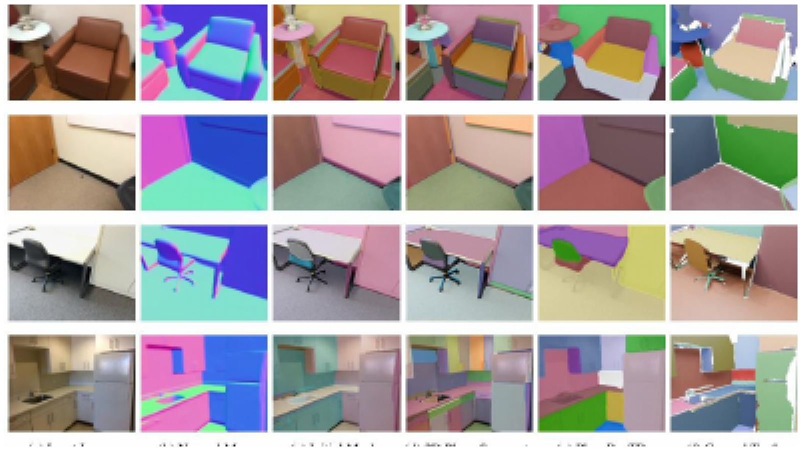
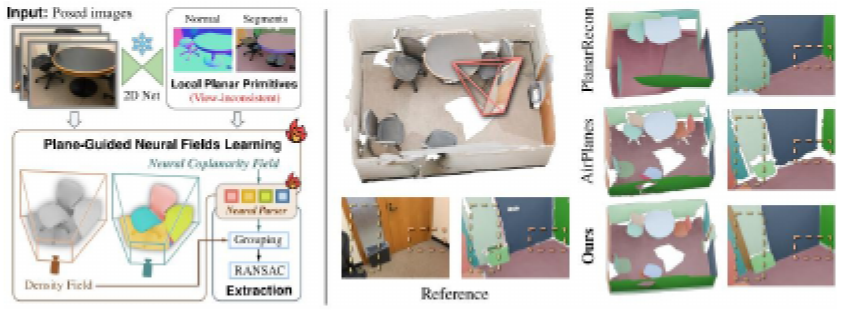
Motivation & Problem Context

3 Why Plane-Guided 3D for Indoor Scenes?



- Clutter and occlusion derail naive reconstructions
  - Textureless walls and repeats confuse cues
  - Under/over-segmentation fractures planar support
- Dense grids miss sparse, structured primitives
  - Geometric regularities (planes) stabilize learning
  - Beyond geometry+RANSAC: model full plane traits

4 NeuralPlane at a Glance



- Multi-view images feed plane-aware neural fields
  - Single-view plane segments seed local primitives
  - Intra-primitive geometry regularizes density field
- Neural Coplanarity Field learns coplanar semantics
  - Produces consistent segmentation across views
  - Outputs structured 3D: cleaner geometry, layouts

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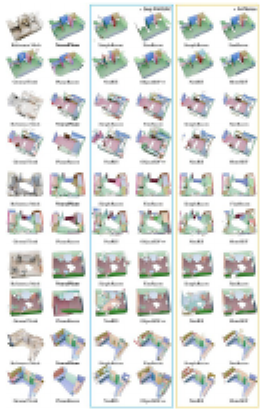
Related Work

3D Positioning vs. Prior 3D Methods

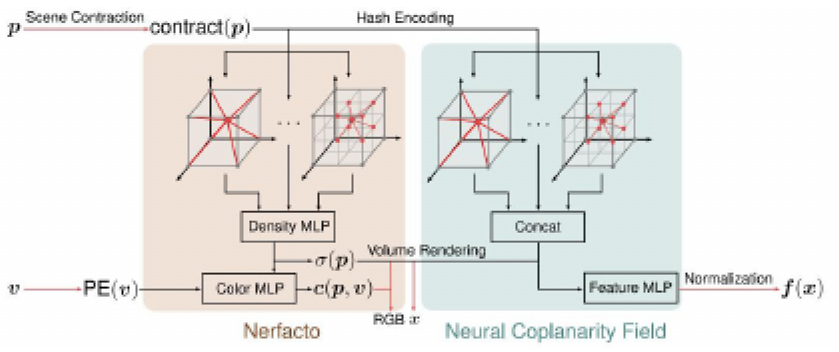


- Traditional MVS struggles to enforce planar regularity.
  - Neural methods add semantics yet lack coplanarity.
  - PlanarRecon fuses planes incrementally; cross-view drift.
- NeuralPlane bridges via plane-guided neural fields.

Method

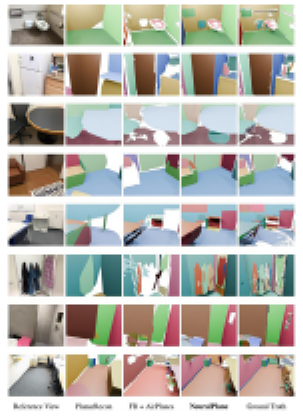
Method — NeuralPlane

8 Plane-Guided Neural Rep. Learning



- Local planar primitives via normals, SAM, K-means
  - Multi-resolution hash encoding feeds MLPs
  - Neural Coplanarity Field predicts coplanarity
- Normal regularization with pseudo-depth supervision
  - Volume rendering fuses density with plane cues
  - Two-branch training: geometry + coplanar semantics

4 Local Primitives to Global Planes



- Cluster coplanar points via DBSCAN + features
  - Leverage semantic prototypes to guide grouping
  - Fit global parametric planes with RANSAC-like
- Enforce multi-view consistency; triangulate meshes

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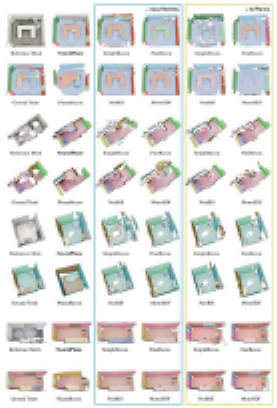
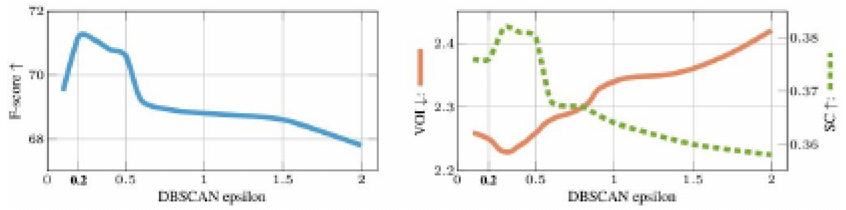
Experiments & Results

3 Evaluation: Fidelity & Robustness



- Evaluated on ScanNetv2 and ScanNet++ indoor scenes.
  - Beats SimpleRecon/MonoSDF on F-score, VOI, SC.
  - Sharper, cleaner planes; aligned semantics in low texture.
- Robust to varying seeds and initialization choices.

12 Ablations: Dim, Epsilon, Prototypes



(a) Geometry (b) Segmentation

- Feature dimensions: F-score saturates as dims grow
  - VOI drops then plateaus with higher feature dims
  - DBSCAN epsilon too high merges planes; F-score falls
- Too low epsilon fragments planes; VOI spikes
  - Prototype granularity trades detail for stability
  - Coplanarity field + parser integration boosts metrics